ENVIRONMENTAL SCIENCES

MODULE 5: WATER POLLUTION

WATER POLLUTION:

- **contamination of water** bodies by harmful substances that degrade water quality
- major cause of waterborne diseases

TYPES OF WATER BODIES AFFECTED BY POLLUTION

1. **SURFACE WATER:**
- includes rivers, lakes, streas, and reservoirs
- pollutants: industrial discharge, agricultural runoff, domestic sewage, urban stormwater runoff

2. **GROUNDWATER**
- **water stored beneath** the Earth's surface in aquifers and underground reservoirs
- pollutants: leakage from septic tanks, agricultural chemicals, waste disposa, landfill leachate

3. **MARINE WATER**
- oceans, seas, coastal areas
- pollutants: river discharge, oil spills, shipping, coastal development

SOURCES OF WATER POLLUTION

A. **POINT SOURCES**
- **specific, identifiable locations** where pollutants are discharged
- generally easier to regulate and control

B. **NON-POINT SOURCES**
- comes from **diffuse sources spread over large areas**
- more difficult to identify and regulate

C. **NATURAL SOURCES:**
- natural processes like volcanic eruptions, natural erosions, etc.

TYPE OF WATER POLLUTANTS

A. **PHYSICAL POLLUTANTS:**
- affect the **appearance** and **physical characteristics** of water

B. **CHEMICAL POLLUTANTS:**
- **substances** that alter <u>chemical composition</u> of water

C. **BIOLOGICAL POLLUTANTS:**
- **microorganisms** that can cause disease

D. **RADIOACTIVE POLLUTANTS:**
- **nuclear power generation**, mining activities, improper disposal

TYPES OF WATER BODIES AFFECTED BY POLLUTION

1. **NUTRIENT POLLUTION:**
- when **excess nutrients** such as <u>nitogen</u> and <u>phosphorus</u> enter water bodies
- causes **eutrophication**, leading to **rapid growth of algae** and **depletion of oxygen in water**

2. **MICROBIAL POLLUTION:**
- when water is **contaminated with pathogenic microorganisms**
- causes cholera, typhoid, desentery

3. **CHEMICAL POLLUTION:**
- results from **industrial and agricultural chemicals**
- may accumulate through **bioaccumulation**

4. **OIL POLLUTION:**
- occurs when **petroleum products** are released
- through **oil spills** or **leaks** from ships and pipelines
- forms a **film on the water surface** that **blocks sunlight** and **oxygen exchange**

5. **THERMAL POLLUTION:**
- when **industries discharge heated water** from power plants
- **increased water temperature** reduces dissolved oxygen levels & disrupts aquatic ecosystems

EFFECTS OF WATER POLLUTION

A. **EFFECTS ON HUMAN HEALTH:**
- waterborne diseases such as cholera and typhoid
- exposure to toxic chemicals
- contaminated drinking water

B. **EFFECTS ON AQUATIC ECOSYSTEMS:**
- fish kills due to oxygen depletion
- destruction of habitatas
- reduction in biodiversity
- disruption of food chain

C. **ENVIRONMENTAL IMPACTS:**
- eutrophication and algal blooms
- bioaccumulation of toxins in food chains
- degradation of wetlands and coastal ecosystems

INDICATORS OF WATER POLLUTION

A. **BIOLOGICAL OXYGEN DEMAND (BOD):**
- measures **organic pollution**
- **amount of oxygen** microorganisms <u>need to decompose organic matter</u>
- indicates sewage or organic pollution

B. **CHEMICAL OXYGEN DEMAND (COD):**
- measures **chemical pollution**
- measures oxygen required to chemicially oxidize pollutants
- indicates **industrial waste** or **chemical contamination**

C. DISSOLVED OXYGEN (DO):
- indicates **water quality** for aquatic life
- **suggests chemical contamination**, **acid rain**, or **industrial waste**
- measures oxygen available for aquatic life

D. TURBIDITY:
- measures **water clarity/ cloudiness**
- indicates **sediments**, **microorganisms**, or **runoff**

E. pH LEVEL:
- measures **acidity** or **alkalinity**
- may indicate **chemical contamination**, **acid rain**, or **industrial waste**

F. COLIFORM BACTERIA COUNT:
- indicates **fecal contamination**
- presence of **microbial contamination**

WATER POLLUTION CONTROL & PREVENTION

A. WASTEWATER TREATMENT:
- removes contaminants **before water is discharged** into the environment
  - PRIMARY TREAMENT – **removal of solids** through **screening** and **sedimentation**
  - SECONDARY TREAMENT – **biological** process to **remove organic matter**
  - TERTIARY TREAMENT – **advanced** processes to **remove nutrients** and **pathogens**

B. POLLUTION PREVENTION STRATEGIES:
- proper industrial waste management
- reduced use of harmful chemicals
- proper disposal household waste

C. WATERSHED MANAGEMENT:
- focuses on **protecting the land areas** that supply water
- reforestation, soil conservation, protection of wetlands, sustainable land-use planning

D. PUBLIC AWARENESS & EDUCATION:
- **public participation**
- environmental education programs, community clean-up, promotion of water conservation

MODULE 6: WATER QUALITY MANAGEMENT & WATER TREATMENT

WATER QUALITY MANAGEMENT:
- refers to the **systematic monitoring**, **protection**, and **improvement** of water resources

PHYSICAL PARAMETERS:
- **physical characteristics** influence the **appearance** and **temperature conditions** of water

TEMPERATURE:
- **affects chemical reactions** and **biological activity** in water

TURBIDITY:
- indicates the **presence of suspended particles**

COLOR:
- may result from **dissolved organic substances** or industrial waste

TOTAL SUSPENDED SOLIDS (TSS):
- particles that **remain suspended** in water

CHEMICAL PARAMETERS
- determine the composition of **dissolved substances**

pH:
- measures the **acidity or alkalinity** of water

DISSOLVED OXYGEN (DO):
- essential for **aquatic life survival**

BIOCHEMICAL OXYGEN DEMAND (BOD):
- indicates the **amount of oxygen** required by microorganisms to **decompose organic matter**

CHEMICAL OXYGEN DEMAND (COD):
- measures the **oxygen needed to oxidize** organic & inorganic substances

NUTRIENTS:
- nitrates and phosphates that can cause euthrophication

HEAVY:
- lead, mercury, asenic, and cadmium

BIOLOGICAL FACTORS:
- measure the presence of living organisms that may affect water safety

WATER QUALITY MONITORING
- fundamental component of water quality management
- involves **systematic sampling**, **testing**, and **analysis** of water conditions

A. WATER SAMPLING:
- collected from water bodies
- must ensure the results accurately represent the water condition

B. LABORATORY ANALYSIS:
- samples are tested in labaratoriess to measure **physical**, **chemical**, and **biological** parameters using **standardized methods**

C. CONTINUOUS MONITORING SYSTEMS:
- **advanced technologies** such as automated sensors and remote monitoring stations **allow real-time tracking** of water quality indicators

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D. WATER QUALITY INDICES:
- combine multiple parameters into a **single numerical vlaue**, simplifying the interpretation of water quality data

WATER QUALITY MANAGEMENT STRATEGIES
- effective management stratigies aim to prevent pollution and maintain acceptable water quality levels
- relies on legal frameworks and policies

A. POLLUTION PREVENTION:
- **preventing** pollution is **more efficient** and **cost-effective** than treating contaminated water

B. WASTEWATER TREATMENT:
- important strategy for **improving water quality before wastewater is discharged**
  - PRIMARY TREAMENT – **removal of solids** through **screening** and **sedimentation**
  - SECONDARY TREAMENT – **biological** process to **remove organic matter**
  - TERTIARY TREAMENT – **advanced** processes to **remove nutrients** and **pathogens**

C. WATERSHED MANAGEMENT:
- protecting the **entire drainage area**

WATER TREATMENT:
- process of **improving the quality of water** to make it suitable for a specific end use

SOURCES OF RAW WATER

A. SURFACE WATER:
- **water collected** from water bodies (rivers, lakes)

B. GROUNDWATER:
- obtained from **wells** and **underground aquifers**
- generally clearer and contains fewer pathogens than surface water
- may contain dissolved minerals which require chemical treatment

C. RAIN WATER:
- **rainwater harvesting systems** collect water from rooftops and other catchment surface
- clean but can still contain dust, microorganisms, and atmospheric polutants

COMMON CONTAMINANTS IN RAW WATER

A. PHYSICAL CONTAMINANTS:
- affects the **apearance and clarity** of water
- increases **turbidity** which can interfere with disinfection process

B. CHEMICAL CONTAMINANTS:
- originate from **natural sources** or **human activities** such as agriculture, industrial discharge, etc.
- toxic at even low concentrations

C. BIOLOGICAL CONTAMINANTS:
- include **microorganisms** that cause **disease**
- **pathogens** are primary cause of waterborne diseases

CONVENTIONAL WATER TREATMENT PROCESS

A. COAGULATION:
- **addition of chemicals** such as aluminum sulfate or ferric chloride to water
- **neutralizes the electrical charges** of suspended particles
- essential for **removing very small particles** that cannot settle naturally

B. FLOCCULATION:
- follows coagulation
- involves **gentle mixing of water**
- encourage particles to collide and **form larger clumps called flocs**, which becomes heavy enough to settle out

C. SEDIMENTATION:
- sedimentation tanks allows heavy flocs **settle at the bottom** due to gravity
- settled particles form **sludge**
- removes a **large portion of suspended solids**

D. FILTRATION:
- water passes through filtration systems to **remove remaining particles**
  - SAND FILTRATION – water flows through **layers of sand** and gravel that trap suspended particles
  - ACTIVATED CARBON FILTRATION – **removes organic compounds**, chlorine, and odors
  - MEMBRANE FILTRATION – membranes with **microscopic pores** remove bacteriia and fine particles

E. DISINFECTION:
- **final step** used to destroy harmful microorganisms that may remain in water
  - CHLORINATION – uses **chlorine** because of its effectiveness and provides **residual protection** in the distribution system
  - ULTRAVIOLET TREATMENT – damages **microbial DNA**, preventing microorganisms from reproducing
  - OZONIZATION – **powerful oxidizing agent** that kills pathogens and removes organic contaminants

ADVANCED WATER TREATMENT TECHNOLOGIES

A. REVERSE OSMOSIS:
- uses **semi-permeable membranes** to remove contaminants
- widely used in **desalination** and **purification systems**

B. ION EXCHANGE:
- **remove unwanted ions** such as calcium and magnesium that causes **water hardness**

- **replaces ions** with less harmful ones like sodium

C. DESALINATION:
- **removes salt** and **minerals** from seawater or brackish water

MODULE 7: AIR POLLUTION

AIR POLLUTION:
- the **presence of harmful substances** in the **atmosphere**

TYPES OF AIR POLLUTANTS

A. PRIMARY POLLUTANTS:
- emmited **directly into th atmosphere** from identifiable sources

1. CARBON MONOXIDE (CO):
- produced from **incomplete combustion** of fuels
- redices the **blood's ability to carry oxygen** – dizziness, headaches, etc

2. SULFUR DIOXIDE (SO2):
- released from **burning coal and oil** in powerplants
- cause **respiratory irritation** and contribute to **acid rain**

3. NITROGEN OXIDES (NO):
- produced from **high temperature combustion** process
- **formation of smog**, **acid rain**, etc.

4. PARTICULATE MATTER (PM):
- **tiny particles** of dust, smoke, and soot **suspended in the air**
- can **penetrate deep into lungs** causing **respiratory and cardiovascular diseases**

5. VOLATILE ORGANIC COMPOUNDS (VOCs):
- emmited from **industrial chemicals**, paints, and fuel evaporation

B. SECONDARY POLLUTANTS:
- formed through **chemical reactions in the atmosphere** involving primary pollutants

C. NATURAL SOURCES:
- natural events like volcanic eruptions releasing sulfur gases, forest fires, etc.

D. ANTHROPOHENIC SOURCES:
- **human activities** are the major contributors to air pollution
- transportation, power generation, industrial, domestic, and agricultural activities,

AIR QUALITY MONITORING
- helps **determine pollution levels** and **identify potential health risks**

A. AIR SAMPLING:
- collected using **specialized instruments** to **measure pollutant concentrations**

B. AIR QUALITY INDEX (AQI)
- **numerical scale** used to communicate how polluted the air and its potential health impacts

MODULE 8: AIR POLLUTION CONTROL METHODS

AIR POLLUTION CONTROL:
- **technologies, strategies, and regulatory measures** to reduce or eliminate harmful pollutants

OBJECTIVES OF AIR POLLUTION CONTROL

- Reduction of pollutant emiisions
- Protection of human health
- Prevention of environmental damage
- Compliance with environmental standards and regulations
- Promotion of sustainable and clean technologies

CLASSIFICATION OF AIR POLLUTION CONTROL METHODS

A. SOURCE CONTROL:
- reduce pollution **before it enters** the atmosphere
- most effective long-term solution because they **prevent emissions** at the souce

B. PROCESS MODIFICATION:
- **changing industrial process** to reduce pollutant formation

C. END-OF-PIPE CONTROL TECHNOLOGIES:
- **installed after** pollution has been generated but before it is released into atmosphere
- capture and neutralize pollutants from **exhaust gases**

PARTICULATE MATTER CONTROL TECHNOLOGIES

A. CYCLONE SEPARATORS:
- remove particles by using **centrifugal force**
- has a **cylindrical chamber** where polluted air spins rapidly
- less efficient for very fine particles

B. FABRIC FILTERS (BAGHOUSE FILTERS):
- use **textile filter bags** to trap particulate matter
- effective for fine particles

C. ELECTROSTATIC PRECIPITATORS (ESP):
- remove particles by using **electrical charges**

- **gas stream** electrically charges particles and are attracted to oppositely charged **metal plates**

D. **WET SCRUBBERS:**
- remove particles by **spraying liquid** into plluted gas streams
- particles **collide with water droplets** and are captured, forming a **slurry** that can be removed

CONTROL OF GASEOUS POLLUTANTS

A. **ABSORPTION SYSTEMS:**
- **dissolving gaseous pollutants** into a liquid solvent
- (e.g.) scrubbing towers use alkaline solutions to remove sulfur dioxide from flue gases in powerplants

B. **ADSORPTION SYSTEMS:**
- capturing pollutants on the **surface of solid materials**, usually **activated carbon**

C. **CATALYCTIC CONVERTERS:**
- widely used in **automobile exhaust systems**
- uses **catalysts** such as platinum, palladium, or rhodium to **convert** harmful gases into less harmful

D. **THERMAL OXIDATION:**
- destroy pollutants by **burning them at high temperature**

CONTROL OF NITROGEN OXIDES (NOx)

A. **SELECTIVE CATALYCTIC REDUCTION (SCR):**
- reduce NOx by **injective ammonia into exhaust gases** in the presence of a **catalyst**
- converts NOx into N and water vapor

B. **SELECTIVE NON-CATALYCTIC REDUCTION (SNCR):**
- also uses **ammonia or urea** but **without a catalyst** at high temperatures